Visit **www.brpaper.com** for downloading previous years question papers of
10th and 12th (PSEB and CBSE), IKPTU,MRSSTU, PSBTE, PANJAB UNIVERSITY,
PUNJABI UNIVERSITY, BFUHS, HPTU, HPSBTE , HARYANA DIPLOMA, MDU HARYANA

Roll No. ☐☐☐☐☐☐☐☐☐☐☐☐ **Total No. of Pages : 02**

Total No. of Questions : 18

B.Tech. (Electrical & Electronics) (2018 Batch) (Sem.–4)

MATHEMATICS-III (PROBABILITY & STATISTICS)

Subject Code : BTAM-302-18

M.Code : 77610

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

SECTION-A

Answer the following in short :

Q1 Define Kurtosis.

Q2 Find the expected number of heads appear, when a coin is tossed twice.

Q3 Write the mean and variance of poisson distribution.

Q4 Write the formula for χ^2 test.

Q5 If a random variable X follows Poisson distribution such that $P(X = 1) = P(X = 2)$, find the mean of distribution.

Q6 Find the probability that 6 out of 10 persons will recover from fever if the probability of recovering from fever is independent and 0.70.

Q7 A problem of mathematics is given to three students A, B and C whose chances of solving it are $\frac{1}{2}, \frac{2}{3}$ and $\frac{3}{4}$ respectively. Find the probability that problem will be solved, if all of them try independently.

Q8 The mean life time of sample of 100 fluorescent light bulbs produced by a company is computed to be 1570 hours with a standard deviation of 120 hours. The company claims that the average life of the bulbs produced by it is 1600 hours. Obtain 95% confidence limits of the mean of the population.

Q9 A random sample of 900 members has a mean 3.4 cms. Can it be regarded as a sample from a large population of mean 3.2 cms and S.D. 2.3 cms?

Q10 Find the mean and variance of the density function $f(x) = ae^{-ax}$, $a > 0$.

BR paper
exams preparatory

SECTION-B

Q11 If the chance that any one of the 10 telephone lines is busy at an instant is 0.2, what is the chance that 5 of the lines are busy? What is the probability that all the lines are busy?

Q12 A sample of 20 items has mean 42 units and S.D. 5 units. Test the hypothesis that, it is a random sample from a normal population with mean 45 units.

Q13 Two random samples are drawn from two normal populations are as follows :

Sample A :	17	27	18	25	27	29	13	17
Sample A :	16	16	20	27	26	25	21	

Test whether the samples are drawn from the same normal population.

Q14 In a normal distribution, 7% of the items are under 35 and 89% are under 63. Find the mean and standard deviation of the distribution.

Q15 Find the value of correlation coefficient between x and y using the following result obtained after performing 15 experiments.

$$\sum x = 100.4,\ \sum x^2 = 750,\ \sum xy = 2055,\ \sum y = 280,\ \sum y^2 = 5690.5$$

SECTION-C

Q16 Obtain the two lines of regression from the following pairs of observations on X and Y: (1, 3), (2, 3), (3, 5), (4, 6), (5, 5), (6, 8), (7, 8), (12, 10). Hence predict the value of Y for X = 3.5.

Q17 Four coins were tossed 200 times. The number of tosses showing 0, 1, 2, 3, 4 heads were found to be as under. Fit a binomial distribution to the data. Also, find the expected frequencies.

No. of heads :	0	1	2	3	4
No. of tosses :	15	35	90	40	20

Q18 Determine the constants a and b by the least squares method such that $y = ax^b$ fits the following data :

***x* :**	1	2	3	4	5
***y* :**	0.5	2	4.5	8	12.5

brpaper.com
brpaper.com